Data in the age of AI

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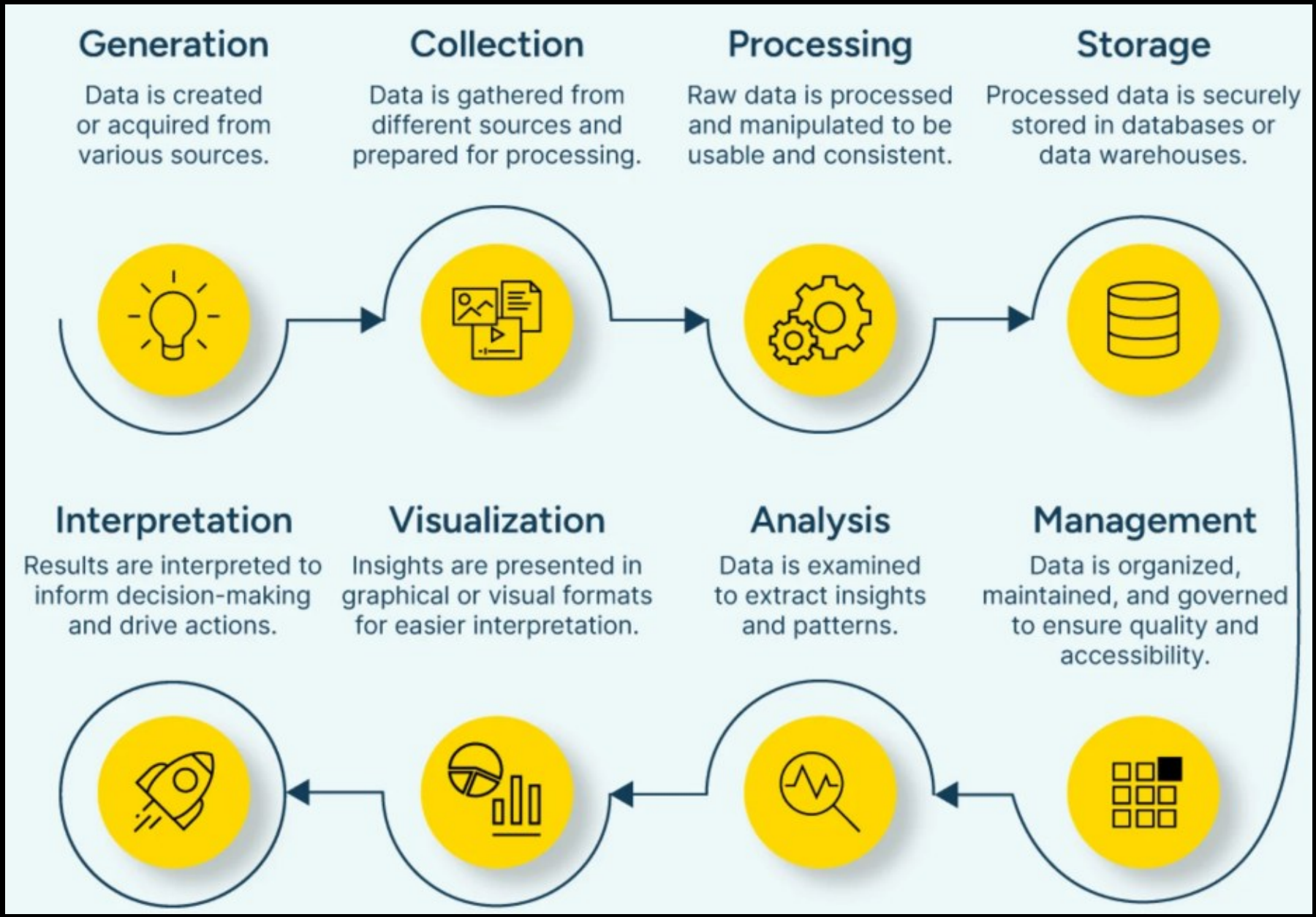
Timeline of information explosion

- 1991: The Internet, or World Wide Web as we know it, became publicly accessible.
- 1995: The Global Positioning System (GPS) reached full operational capability.
- 1997: The term “Big Data” was used in academic research.
- 1998: Google was founded by Larry Page and Sergey Brin.
- 2001: Wikipedia was launched as a free online encyclopaedia.
- 2002: Bluetooth became a globally recognized wireless technology integrated into the first mass-market laptops and mobile phones.
- 2008: The number of devices connected to the Internet exceeded the global population.
- 2015: Cloud services like Microsoft Azure and Google Cloud Platform became widely recognized as major enterprise cloud providers.
- 2020: Global data creation reached approximately 64 zettabytes (10^{21} bytes).

The traditional data lifecycle



Traditional data are evaluated in relation to their capacity to support human decision-making processes.

The rise of AI introduces a fundamental shift: data must now be fit not only for human interpretation but also for **algorithmic processing and training machine learning models.**

AI introduces **dynamic quality requirements**. Unlike static reporting or analysis, where data quality can be assessed once and maintained, machine learning models require **continuous monitoring of data quality throughout their operational lifecycle.**

Generative AI creates **synthetic data** (artificial generated information that mirrors the statistical patterns of real-world data without containing private information).

Big Data

Big data refers to a set of **large amounts of varied and complex information** that represents a great challenge for storage, access, analysis, and processing.

> "Big data isn't just big — it's fast, messy, diverse, uncertain, valuable, and constantly changing."

Big data comprises six characteristics, commonly called 6v's: **volume, value, velocity, variety, veracity, and variability.**

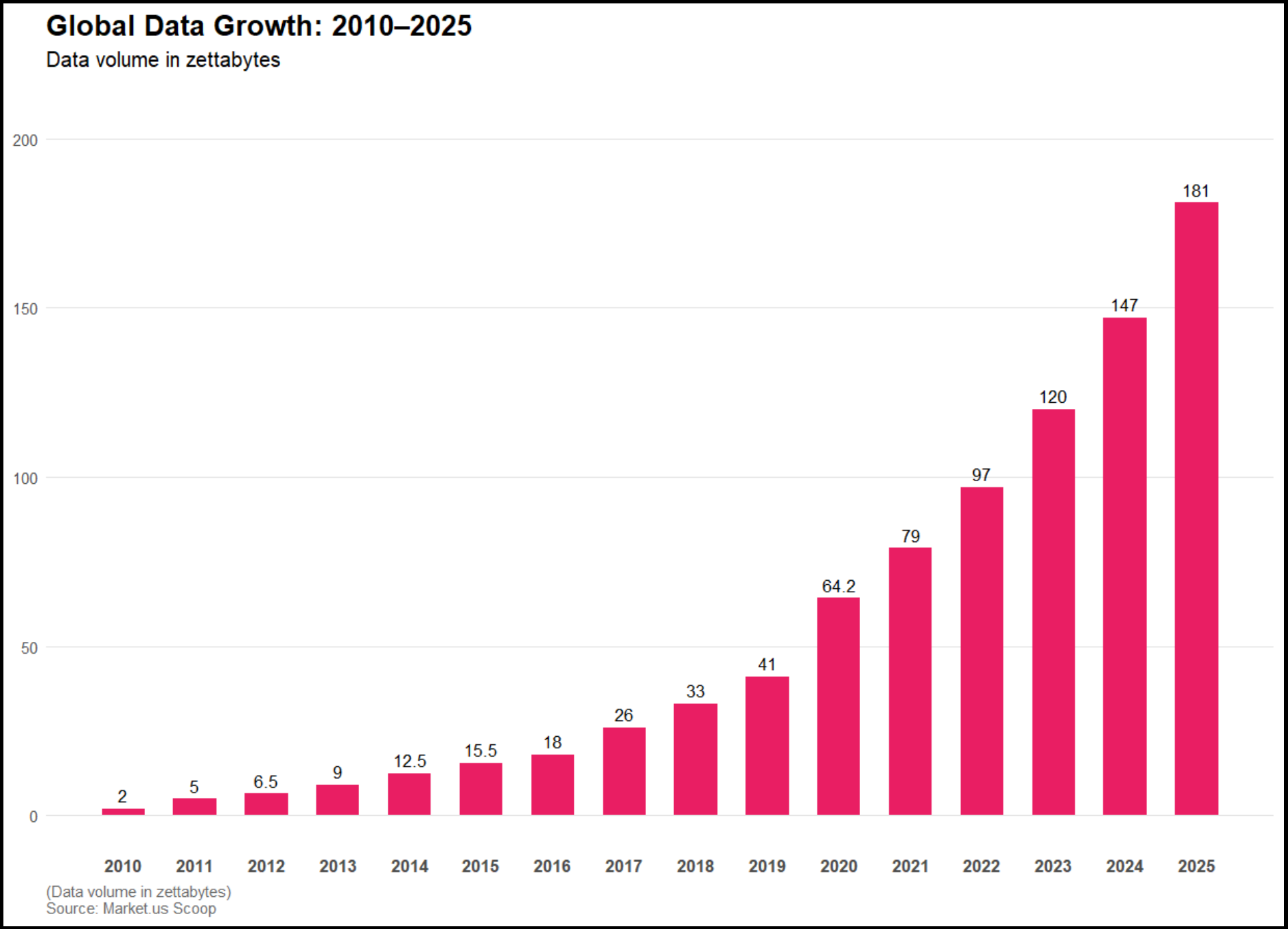
1. Volume

Volume represents the large and growing amount of data generated continuously, reaching magnitudes such as petabyte, exabyte, and zettabyte.

More generally, it indicates volumes that are too large and complex to be handled with conventional data storage and processing technologies.

When data volumes gets large or when the data complexity increases, analytics run times can grow to the point that they take longer to compute than the operational constraints will allow.

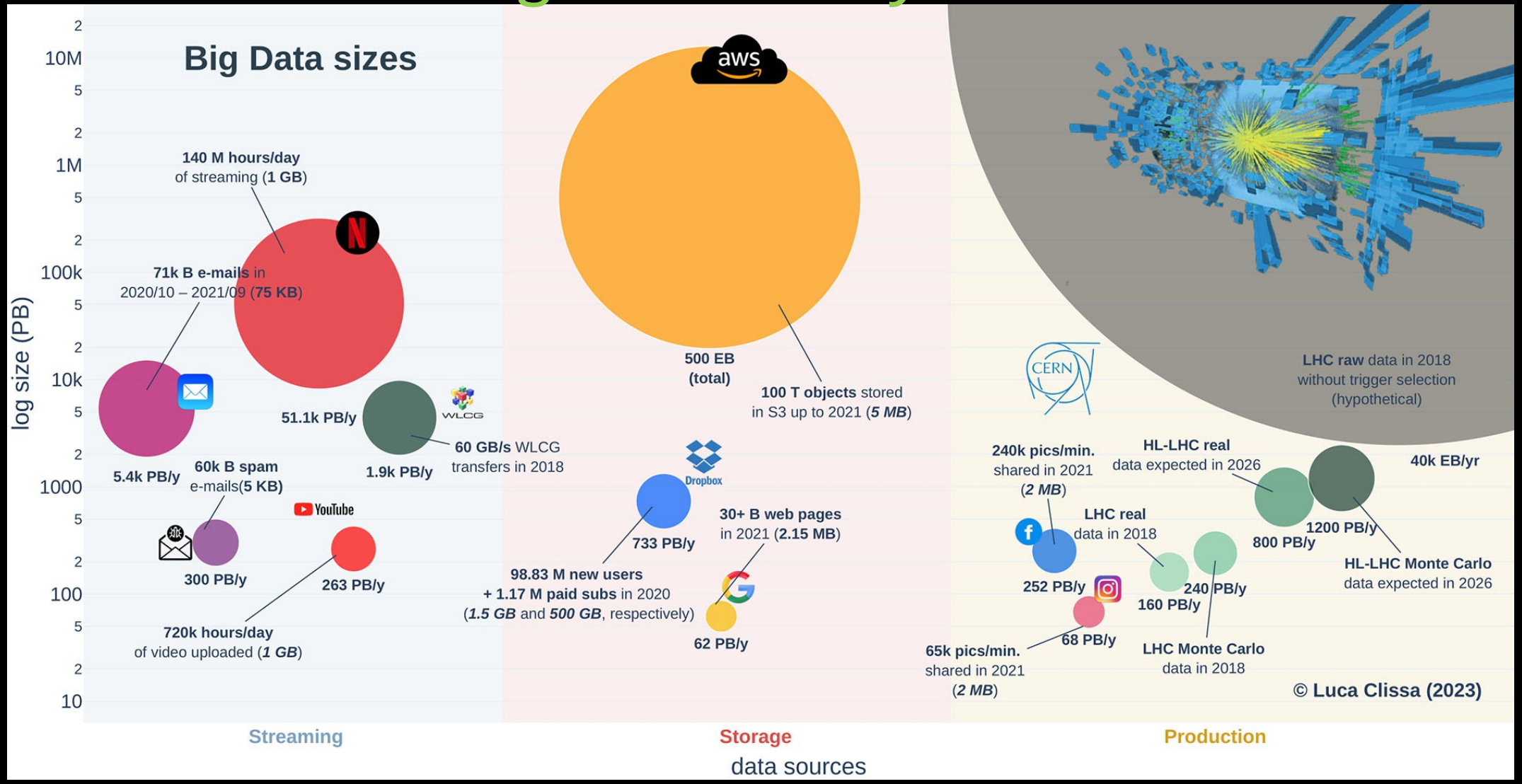
Unit	Value (1 byte can represent 256 different values)	Use case
Megabyte	1,000 kilobytes	A digital copy of the **Iliad** or the **Odyssey.**
Gigabyte	1,000 megabytes	Roughly 230 songs; the complete discography of **Maria Callas.**
Terabyte	1,000 gigabytes	500 hours of movies; every **Eurovision** performance since 1956.
Petabyte	1,000 terabytes	High-resolution photos of every artifact in the **Acropolis Museum**.
Exabyte	1,000 petabytes	The **National Library of Greece,** digitized 3,000 times over.
Zettabyte	1,000 exabytes	36,000 years of HD video
Yottabyte	1,000 zettabytes	A data-center the size of **Crete**


Global Data Growth: 2010–2025
Data volume in zettabytes
200
150
100
50
0
2
5
6.5
9
12.5
15.5
18
26
33
41
64.2
79
97
120
147
181
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
(Data volume in zettabytes)
Source: Market.us Scoop

Sources of very high-volume data

Category	Source Examples	Notes
Web & Social	Facebook, Twitter, Instagram, YouTube	Unstructured text, images, video; often high-volume
Biometric Data	Fitness trackers, genetic tests, health sensors	Personal and health-related measurements
Transactional Systems	Point of Sale (POS), e-commerce systems	Structured transactional records
Internet of Things (IoT)	Smart devices, RFID/ID tags, sensors	Machine-generated sensor data
Cloud Systems	SaaS applications like Salesforce, cloud storage	Business application and collaboration data
Corporate Databases & Spreadsheets	SQL/NoSQL databases, Excel files	Internal structured data repositories

Big Data Players



Orders of magnitude involved in different data sources for several big data players. The area of each bubble represents the amount of data streamed, hosted or generated. The accompanying text annotations emphasize the crucial factors considered in the estimation process. Average per-unit sizes are indicated in parentheses, where italic denotes measures derived from reasonable assumptions due to the absence of available references. https://doi.org/10.3389/fdata.2023.1271639

2. Variety

Variety refers to the various **types of data**, which can be:

- structured, that is, readily classified and analyzed (e.g., tables, databases)
- unstructured, which are not organized according to a predefined model of information (e.g., text notes, images, video, audio)
- semi-structured, which are similar to a mixture of the other two types (e.g., emails)

3. Velocity

Velocity refers to the **speed** at which data is generated, collected, and processed (e.g., real-time streaming data, stock trades, IoT sensors, real-time analysis of credit card swipes to detect fraud instantly).

4. Value

The **usefulness** of data **after** it's been processed and analyzed. Raw data alone means nothing — the goal is to extract meaningful insights that drive decisions.

5. Variability

Variability is defined as **the changes** that information can present with time, location, and among different operators (e.g., seasonal trends, spikes in usage, changing customer behavior).

The "Black Friday" Effect: An e-commerce site might process 1,000 transactions per minute normally, but 1,000,000 per minute on a holiday.

Viral Events: A sudden news story or a viral post can cause a massive, unpredictable spike in social media data that drops off just as fast.

6. Veracity

This refers to dimensions of the **quality of the information**. Big data can be noisy, incomplete, biased, or inconsistent.
If the data is biased or incorrect, the resulting insights will be flawed (the "Garbage In, Garbage Out" principle).

ISO 8000 defines quality data as “portable data that meets stated requirements”. Data must be separable from a software application (portable) and comply with explicit, unambiguous requirements set by the organization, the customer, or the relevant stakeholders.

Core dimensions of data quality (before AI)

Dimension	Definition	Practical Example	Measurement Approach
Accuracy	The degree to which data correctly describes the real-world information.	Patient's recorded blood pressure matches the actual measurement.	Comparison against an authoritative source or ground truth.
Completeness	The extent to which all required data is present.	The customer database contains contact details for all clients.	Ratio of available values to required values; percentage of missing fields.
Consistency	Absence of contradictions within and across datasets.	A patient's birthdate is consistent across both electronic health records and insurance records.	Cross-field and cross-database validation checks.
Timeliness	The degree to which data is up to date and available when needed.	Stock market prices updated in real time.	Lag time between data generation and availability for use.
Validity	Degree to which data conforms to defined formats, rules, or ranges.	Postal codes follow the official national standard.	Validation rules, format checks, and range constraints.
Relevance	Appropriateness of data for the intended use.	Including clinical trial data when evaluating a new treatment.	Expert judgment; alignment with analytical or decision-making needs.
Uniqueness	The degree to which data is free of duplicate records.	Each patient has a single unique medical record number.	Duplicate detection and record linkage algorithms.

https://doi.org/10.3390/data10120201

AI systems impose specific requirements

- **AI-Readiness:** Data must be structured, annotated, and formatted in a way that allows algorithms to process it efficiently. This includes requirements for standardized schemas, consistent labeling in supervised learning contexts, and appropriate representation of features for model input.

- **Algorithmic Fairness:** Data quality must explicitly account for representation and balance across demographic groups, domains, or categories to ensure fairness. Biased training data perpetuate and amplify discrimination in algorithmic decision-making across sectors such as healthcare, criminal justice, and finance.

  NOTE: *The "Gemini Paradox": Correcting for systemic bias in training data (under-representation) can inadvertently create new biases (historical inaccuracy) if context is ignored.*

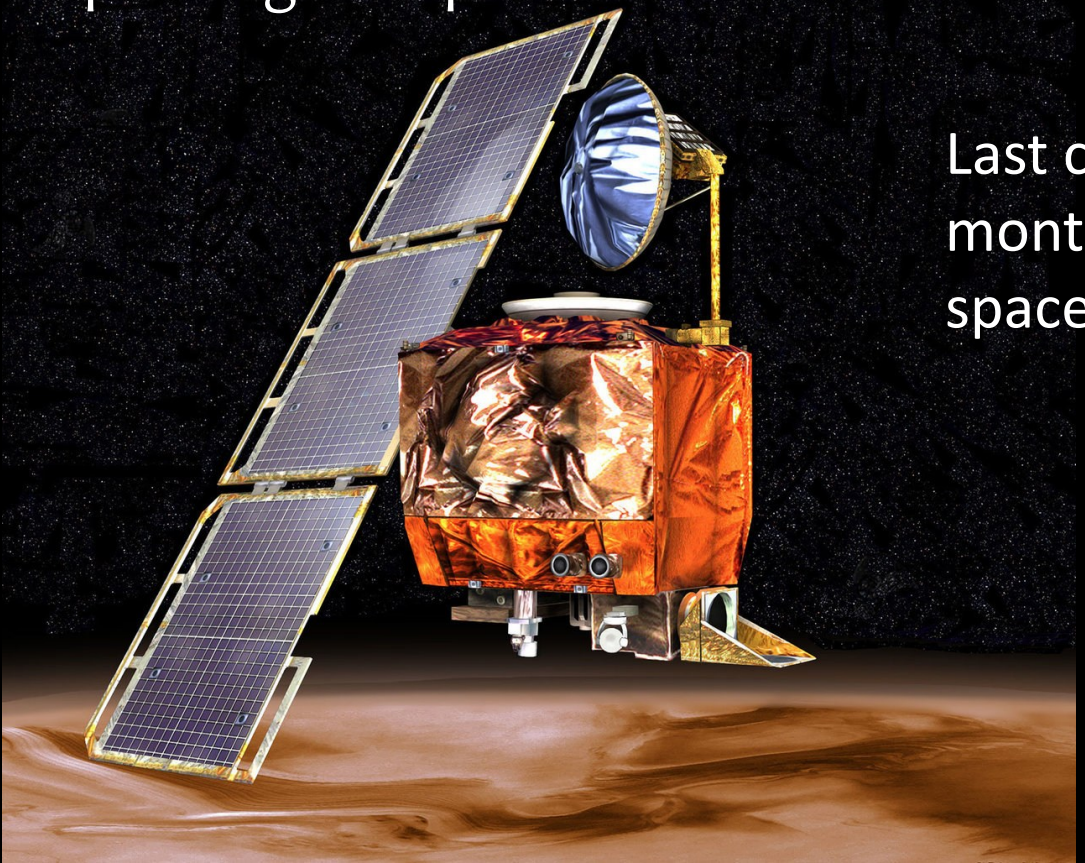
- **Model Explainability and Traceability:** High-quality data must support interpretability requirements. Provenance, lineage, and metadata become critical not merely for reproducibility but for understanding how algorithmic decisions are derived—essential for regulatory compliance and ethical accountability.

AI-specific requirements challenge the traditional separation between intrinsic data properties (accuracy, completeness, and consistency) and contextual dimensions (relevance and timeliness).

In AI contexts, what constitutes “accurate” or “complete” data depends fundamentally on the **model architecture, training objectives, and deployment context**. A dataset may be adequate for descriptive analytics yet wholly insufficient for predictive modeling if it lacks temporal coverage, feature diversity, or a sufficiently large sample size for robust generalization.

NASA’s Mars Climate Orbiter (example of failure)

Designed to study Mars from orbit, the Mars Climate Orbiter mission failed due to a mismatch between two software systems using different measurement units. The ground-based navigation software produced thruster data in pound-force seconds, while the spacecraft’s onboard system expected the metric unit newton-seconds. Because the values were not converted correctly, trajectory calculations became inaccurate, gradually pushing the spacecraft closer to Mars than planned.



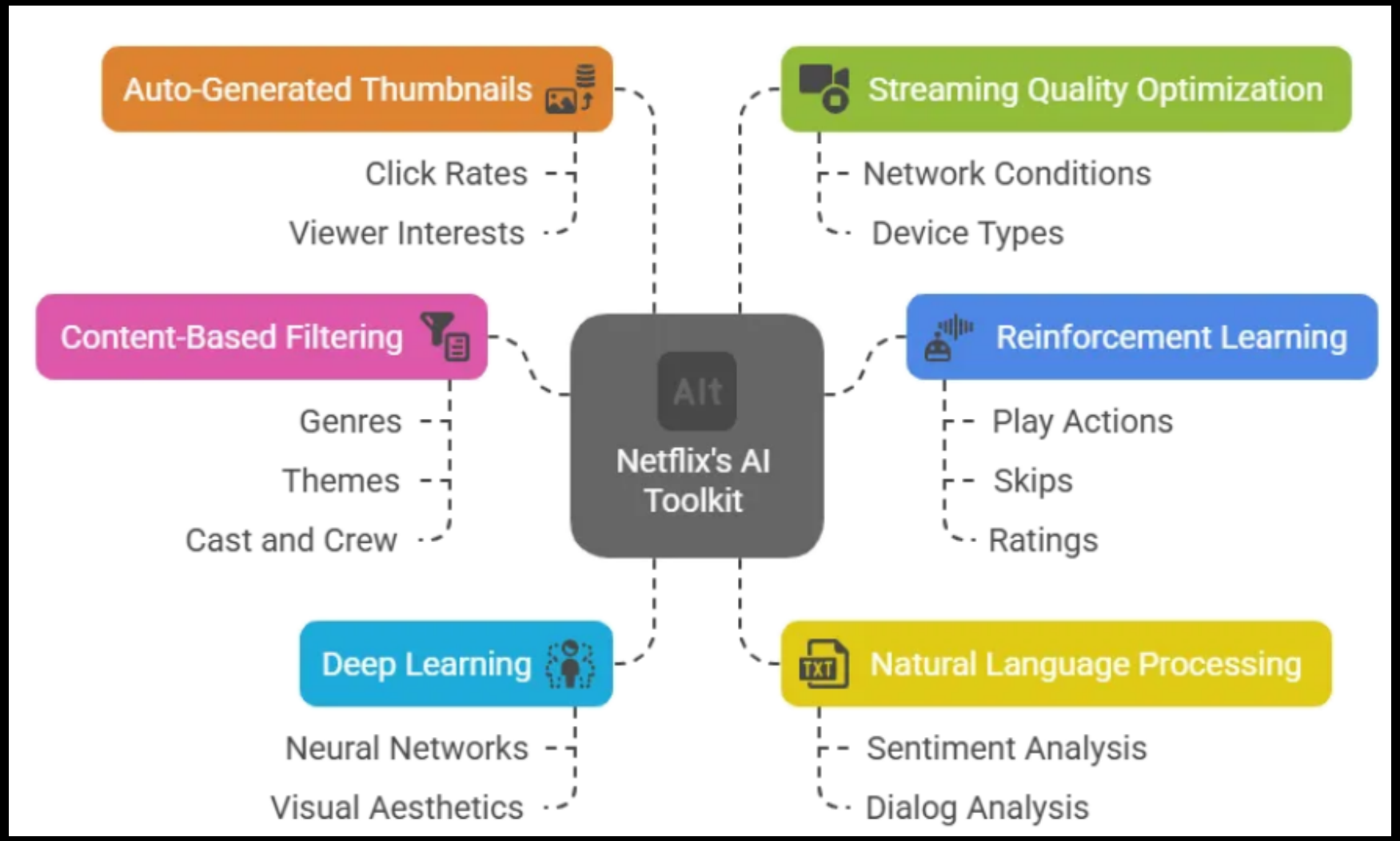
Last contact with the spacecraft was on Sept. 23, 1999, nine months after launch, and an investigation found that the spacecraft burned up in Mars' atmosphere.

The **Mars Climate Orbiter mission cost about $327.6 million** in total.

https://science.nasa.gov/mission/mars-climate-orbiter/

The Netflix Recommender System (example of success)

Leveraging high-quality behavioral data for personalization



For example, if a user likes movies from a certain director or certain actors then content-based filtering takes into consideration all that metadata. Netflix also utilises a kind of algorithm to focus on the top rankings of individual users based on their viewing history as well as their feedback.

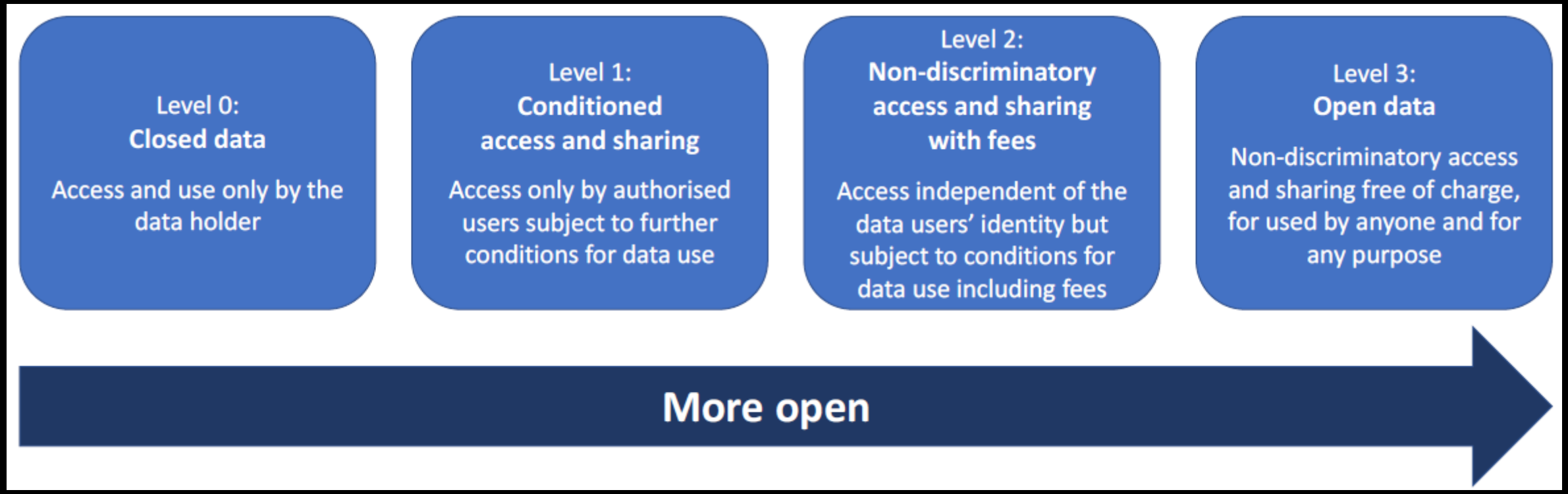
What is Sensitive data

Sensitive data is data that **must be protected** against unwanted disclosure. Access to sensitive data should be safeguarded. Protection of sensitive data may be required for legal or ethical reasons, for issues pertaining to personal privacy, or for proprietary considerations.

Examples of sensitive data are:

- **Personal data:** Identifiers such as names or identification numbers, as well as physical, physiological, genetic, mental, economic, cultural, or social characteristics. This category also includes location data obtained from GPS devices or mobile phones.
- **Confidential data**: Trade secrets, investigation-related information, and data protected by intellectual property rights.
- **Security-related data:** Passwords, financial information, and information related to national security or military activities.
- **Combined datasets:** Different datasets that, when combined, may reveal sensitive or personally identifiable information.
- **Biological data**: Data concerning endangered plant or animal species whose survival depends on protecting their location information (e.g., within biodiversity research communities).
- **Sensitive metadata:** Metadata that may contain or reveal personal or sensitive information.

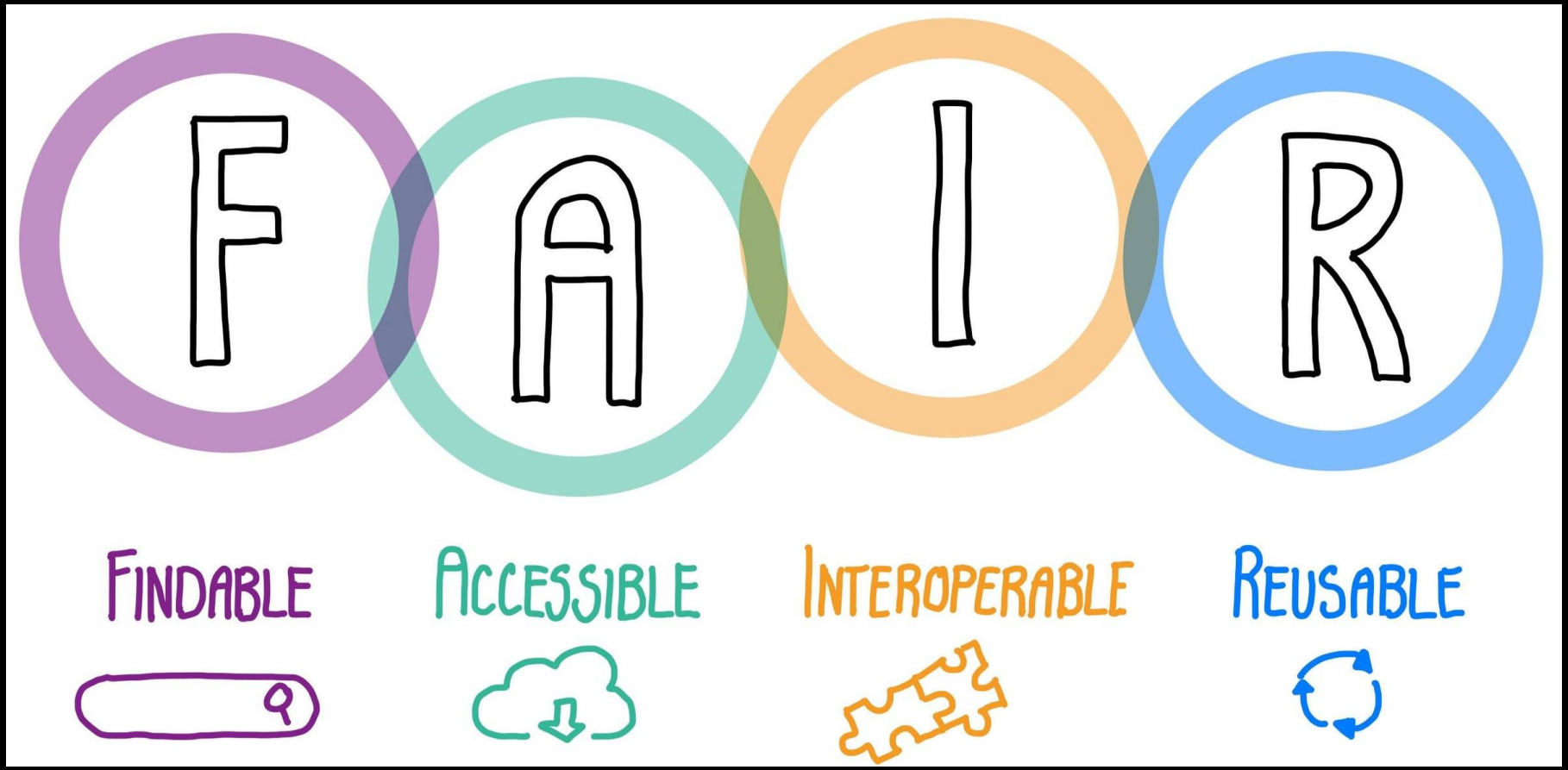
The degrees of data openness



https://doi.org/10.1787/9789264229358-en

Using LLMs with Public and Restricted Data

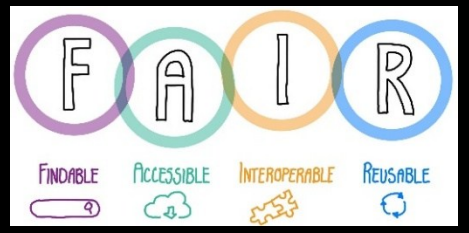
Data type/ AI Context	Public/online LLM (e.g., ChatGPT) that retains data for future training	Online LLM that does not retain data (e.g., Maizey)	"Homegrown" LLM on isolated machines
Public-use	Never okay	Okay with permission, assuming no identification	Okay with permission, assuming no identification
Restricted-use	Never okay	Never okay	Okay with permission, assuming data and LLM are both in environment stipulated in data security plan



The FAIR Principles put specific emphasis on enhancing the ability of machines to automatically find and use the data, in addition to supporting its reuse by individuals.

FAIR principles for data

In AI, FAIR data ensures that training sets are high-quality and machine-readable.



- **Findable:** Data and metadata should be easy to locate for both humans and machines. This involves assigning globally unique and persistent identifiers, providing rich metadata, and registering or indexing data in searchable resources.
- **Accessible:** Once found, data should be retrievable using standardized protocols. This includes ensuring that metadata remains accessible even if the data itself is no longer available.
- **Interoperable:** Data should be compatible with other datasets and tools. This requires using standardized vocabularies and formats, facilitating integration with other data sources.
- **Reusable:** Data should be well-described and documented to allow for replication and further use. This entails clear licensing, detailed provenance information, and adherence to community standards.

https://www.go-fair.org/fair-principles/